Original Write histories and an interacting source-field action

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Abstract. Original signed Write histories supply phase and action data that can be received into a finite pair source coupled to a two-polarization quadrupole field. This chapter keeps the original rules, the receiving laws, and their derived consequences explicit. A four-component source yields six driven spectral lanes and four unforced states. For a declared coherent hard-field preparation in each lane, a weighted continuum comparison bounds the actual source-field correlator. All six instantaneous phase rates remain positive and below the bare gap throughout one bare cycle. The resulting phase coordinate reparameterizes the complete joint action, retaining the even source energy, both counterterms, and the Berry one-form.

1. The physical question and its inputs

An original Write is a signed event in an ordered source history. A receiving field observes specified combinations of that source. Chapter I asks how the original phase incidence, a complete finite source, and a declared field law combine into an interacting action with a usable local phase clock.

Three layers carry different authority. **Original rules** supply the signed Writes, lifted phases, retained tensor map, and source histories. **Receiving laws** specify the quadrupole bulk field, its quantization, the boundary incidence, and the preparation. **Derived compositions** include the source spectrum, conserved charge, actual clock equation, and certified phase reparameterization. An exact composition keeps its inputs identifiable. [1-4]

The construction uses the full source space $\mathbb{C}^4 \otimes \mathbb{C}^4$ and both real field polarizations. All field frequencies and photon sectors belong to the model. The six clock certificates concern six separately prepared lanes. Each certificate carries its own signed root difference, coupling, and hard-branch preparation.

The numerical example receives the unit-strength source chart, the retained boundary radius, and a capacity-normalized model time. A physical exchange strength, the reference capacity and its role, the Write-to-seconds assignment, and the incidence between the source sphere and field boundary remain explicit physical inputs. No inverse-seconds rate is selected by the normalized calculation.

2. Lifted phases and history

For lifted endpoint phases φ_a, φ_b and partner phase φ_p, define

$$\gamma_W = \phi_a + \phi_b - 2\phi_p,$$
$$\delta = \phi_a - \phi_b, \qquad \chi = \phi_p. \qquad (1)$$

The invertible incidence is

$$\phi_a = \frac{1}{2}(\gamma_W + \delta) + \chi,$$
$$\phi_b = \frac{1}{2}(\gamma_W - \delta) + \chi. \qquad (2)$$

On the legal piecewise active Write paths, the original action ledger is

$$\omega_{\mathrm{Write}} = \hbar(d\gamma_W + 3d\chi). \qquad (3)$$

The factor three follows by substituting (2) into the sum of the three phase differentials. The retained tensor orientation obeys $\psi = \psi_0 - \gamma_W/2$. The common phase χ remains part of the action account even when a receiving observable is insensitive to it. [1]

Lifted cumulative phase, a wrapped endpoint, and an ordered event word retain different information. The completed H02 and H03 histories reach the same source endpoint with zero total signed phase action, yet have positive durations approximately 0.3156476595 and 0.3181630929 in their stated source calibration. Their difference is a difference of positive durations. [1, §6]

The interacting clock phase θ introduced below remains distinct from γ_W and χ.

3. A complete four-component source

The original placement codes 46501, 46525, 49061, and 81829 define the retained signed boundary construction. Its 4 × 78 incidence matrix B gives T = BB*, and the retained signed response supplies K. The receiving family is $Q(g_H) = T + g_H K$. [1, 4]

At the inherited diagnostic strength $g_H = 1$, Q is the following real symmetric matrix:

	0	1	2	3
0	756	104	40	0
1	104	2100	0	0
2	40	0	756	40
3	0	0	40	756

With x = q − 756, its characteristic polynomial is

$$p(x) = x^4 - 1344x^3 - 14016x^2$$
$$+4300800x + 17305600. \qquad (4)$$

The ordered simple eigenvalues have the following approximate displays. The calculation uses isolated exact roots and native enclosures, rather than these rounded values.

a	q_a
0	697.330492591257
1	752.008183657655
2	810.654350286272
3	2108.006973464815

The source response ledger and the physical strength g_H remain distinct from this completed matrix example. The source record retains T, K, the general polynomial, and the spectral calibration weights. [4, 6]

3.1 Six lanes and four diagonal states

Let $Q_L = Q \otimes I$, $Q_R = I \otimes Q$, and S = Swap. The complete pair operators are

$$H_{\text{pair}} = S + Q_L + Q_R + Q_L Q_R,$$
$$D = Q_L - Q_R, \qquad C = iSD. \qquad (5)$$

For a < b, the ordered lane basis is $(|q_a q_b\rangle, |q_b q_a\rangle)$. Set

$$d_{ab} = q_a - q_b < 0,$$
$$e_{ab} = q_a + q_b + q_a q_b. \qquad (6)$$

On that two-dimensional lane,

$$S = \sigma_x, \quad D = d_{ab}\sigma_z, \quad C = d_{ab}\sigma_y,$$
$$H_{\text{pair}} = e_{ab} I + \sigma_x. \qquad (7)$$

Thus the source eigenenergies are $e_{ab} \pm 1$, while $D^2 = C^2 = d_{ab}^2 I$. The sign of d_{ab} remains in the interaction and correlated clock equation. It is retained when positive energy estimates use d_{ab}^2.

The four diagonal states $|q_a q_a\rangle$ have D = C = 0 and pair energy $(1 + q_a)^2$. Together with the six two-state lanes, they exhaust all sixteen source states.

Writing $N_s = (I + S)/2$ gives $H_{\text{pair}} = 2N_s + K_{\text{pair}}$, where $K_{\text{pair}} = Q_L + Q_R + Q_L Q_R - I$ commutes with the source currents. This even remainder is scalar within a lane and varies across the full source. It remains distinct from the independent original common phase χ.

3.2 Observable maps and retained complements

The earlier full18 tensor representation and this sixteen-state pair source are distinct objects. The retained endpoint-to-tensor construction has tensor Gram $32I_2$ and pair-observable Gram $11062272I_2$. Its observable composition reconstructs two tensor coordinates, while a retained complementary map reconstructs the other source coordinates. All sixteen joint C_4 character sectors remain present. [1, §2.2]

That composition preserves the difference between an observable map and a complete Hilbert action. The literal endpoint D/C action comparison retains exact residual Gram entry 54272. The full18 complement and that residual travel with the source construction; the lane clock below uses the explicit operators (5), rather than identifying these distinct state spaces.

4. Source energy and the time chart

The original calibration vector $|e_0e_0\rangle$ is a placement-basis state. Since $Q_{00} = 756$, its complete pair-energy expectation is

$$E_{\mathrm{ref}} = \langle e_0e_0|H_{\mathrm{pair}}|e_0e_0\rangle$$

$$E_{\mathrm{ref}} = (1 + 756)^2 = 573049. \qquad (8)$$

This expectation fixes the declared capacity normalization. It is not an eigenenergy. The spectral populations reconstruct (8), while the original pure-state density retains its coherences. Its D and C expectations vanish; its nonzero energy variance is 14384109824. It therefore supplies a different preparation from the interacting clock state below. [4, 6]

The spectral calibration weights have an exact resolvent expression. For $x_a = q_a - 756$ and p from (4),

$$p_a = \frac{(x_a - 1344)(x_a^2 - 1600)}{p'(x_a)}.$$

$$\sum_a p_a = 1, \qquad \sum_a p_a q_a = 756,$$

$$\sum_a p_a q_a^2 = 583952.$$

The four diagonal populations p^2_a and six pair populations $2p_ap_b$ reconstruct the block-diagonal energy (8). This use of populations retains the coherences in the original pure state. [6]

For action unit I_*, physical time unit t_u, Write time t_{SW}, and reference capacity λ_{ref}, the receiving coefficient is

$$\varepsilon = \frac{\hbar}{I_*}\frac{t_u}{t_{\mathrm{SW}}}\frac{\lambda_{\mathrm{ref}}}{E_{\mathrm{ref}}}. \qquad (9)$$

The normalized chart $I_* = \hbar$ and $t_u = t_{SW}/\lambda_{ref}$ gives

$$\varepsilon = \frac{1}{573049}, \qquad \Delta = 2\varepsilon. \qquad (10)$$

We use $\tau = t_{phys}/t_u$ throughout. A displayed angular rate ν is per unit τ. Its conditional physical rate is $(\lambda_{ref}/t_{SW})\nu$ in this chart. The source-energy normalization and the signed phase-current gap have separate roles.

5. The complete interacting action

Let Ψ be a normalized joint source-field state. The model-time action is

$$\mathcal{S} = I_* \int d\tau\,[\mathcal{B}_\tau - \langle H\rangle],$$

$$\mathcal{B}_\tau = \frac{i}{2}(\langle\Psi|\dot\Psi\rangle - \langle\dot\Psi|\Psi\rangle). \qquad (11)$$

For continuum trajectories, the derivative and energy pairings use the form domain and its dual, defined in (18). The smooth-core variational expression extends through the closed Hamiltonian form. The full form is

$$H = \varepsilon H_{\mathrm{pair}} + H_f - \varepsilon(D\phi_+ + C\phi_\times)$$

$$+\varepsilon^2\mu(D^2 + C^2), \qquad \mu = \frac{1}{4R}. \qquad (12)$$

Here φ denotes the boundary field in the retained receiving normalization. Equation (12) contains the entire joint Hamiltonian; the source and field actions are already included. In particular, $\langle D\varphi_+\rangle$ and $\langle C\varphi_\times\rangle$ remain joint correlations.

On one fixed lane, put $g = \varepsilon d_{ab}$ and $e = e_{ab}$. Then

$$H = \varepsilon eI + \varepsilon S + K,$$

$$K = H_f - g(\sigma_z\phi_+ + \sigma_y\phi_\times) + cI,$$

$$c = 2g^2\mu = \frac{g^2}{2R}. \qquad (13)$$

Each real polarization contributes $g^2/(4R)$. Both contributions and the even energy εe remain in the action. The positive field block is the square form

$$q_K[\Psi] = \sum_{\alpha = +,\,\times}\int_0^\infty d\omega$$

$$\left\| \sqrt{\omega}\,a_\alpha(\omega)\Psi - \frac{g\sigma_\alpha u_R^*(\omega)}{\sqrt{\omega}}\Psi \right\|^2,$$

where $\sigma_+ = \sigma_z$ and $\sigma_\times = \sigma_y$. The energy-annihilation map and its source shifts act in the same direct-integral space. This defines $K \geq 0$ without requiring the two shifted maps to commute.

Variation of (11) yields joint Schrödinger evolution, with a normalization multiplier contributing a common state phase. [1, §3; 5, 6]

5.1 Boundary and field normalization

The receiving free-field Lagrangian is

$$L_f = \frac{\kappa}{4} \sum_{a=+,\times} \int_0^\infty dr\, \mathcal{L}_a,$$

$$\mathcal{L}_a = r^2 \dot{\varphi}_a^2 - r^2 (\partial_r \varphi_a)^2 - 6\varphi_a^2. \qquad (14)$$

For boundary coordinate $x = R\phi(R)$, the regular static interior is xr^2/R^3 and the exterior is xR^2/r^3. Their gradient-plus-angular integrals are $2x^2/R$ and $3x^2/R$. Matching the sum to the literal tensor metric $B_{TT} = 2I_2$ gives

$$\frac{5\kappa}{4} = 1, \qquad \kappa = \frac{4}{5}. \qquad (15)$$

With radial measure r^2dr, the canonical momentum is $p_a = \kappa\dot{\phi}_a/2$. With measure dr, the same variable is the density $\Pi_a = r^2 p_a$. The boundary source $\varepsilon J_a \phi_a(R)$, $J_+ = D$ and $J_\times = C$, retains the jump

$$[\partial_r \varphi_a]_{R^-}^{R^+} = -\frac{2\varepsilon J_a}{\kappa R^2}. \qquad (16)$$

Choose the real radial representative $u_R(\omega) = \sqrt{[2\omega/(\kappa\pi)]}\, j_2(\omega R)$. This fixes the conjugation convention in the square form above. The quantized boundary kernel is

$$|u_R(\omega)|^2 = \frac{2\omega j_2(\omega R)^2}{\kappa\pi},$$

$$\mu = \int_0^\infty \frac{|u_R(\omega)|^2}{\omega}\, d\omega = \frac{1}{4R}. \qquad (17)$$

The retained boundary radius is $R \approx 0.163544171326059$. The static completion pays $\varepsilon^2 J^2/(4R)$ per current. The static boundary completion makes this payment explicit:

$$\frac{x^2 - \varepsilon J x + \varepsilon^2 J^2/4}{R} = \frac{(x - \varepsilon J/2)^2}{R}.$$

Integrating the radial field equation across the boundary gives (16), with the retained orientation R^- to R^+. The two static pieces therefore fix both the normalization and the boundary sign. Noncommuting source shifts remain in the positive square form; a simultaneous shifted vacuum is not assumed.

5.2 The full continuum and signed charge

Take the bosonic Fock space over $L^2((0,\infty),d\omega;\mathbb{C}^2)$, with $H_f = d\Gamma(\omega)$. The common energy form domain is

$$Q_f = D(H_f^{1/2}), \qquad A = I + H_f. \qquad (18)$$

The weighted annihilation estimate

$$\|a(u_R)\eta\| \le \sqrt{\mu}\, \|H_{f,a}^{1/2}\eta\| \qquad (19)$$

constructs the field pairings on this domain by frequency-cutoff convergence. Creation is defined by the adjoint form pairing. An unweighted L^2 norm of u_R is unnecessary.

For a bounded source current J, the field form satisfies $|q_{J\varphi}[\eta]| \le 2\sqrt{\mu}\, \|J\eta\|\, \|H_{f,a}{}^{1/2}\eta\|$. Young's inequality bounds the full interaction by any positive multiple of the free energy plus a finite multiple of $\|\eta\|^2$.

The finite source therefore makes the interaction infinitesimally bounded relative to the free energy form. The closed semibounded form (12) defines a self-adjoint Hamiltonian with form domain Q_f. Source observables preserve that domain; the form Ehrenfest equation makes their expectations continuously differentiable. Evolution preserves Q_f and is continuous in its energy norm; its derivative exists in $Q_f{}^*$. Taking the expectation difference quotient in this dual pairing gives the form Ehrenfest identity. The resulting commutator forms are continuous in time. This establishes the pointwise regularity used in (22), including for preparations in Q_f. [6]

Both circular polarizations and all four circular interaction words are retained. With $L = (D + iC)/\sqrt{2}$, those words are $La^\dagger_+$, $L^\dagger a_+$, $L^\dagger a^\dagger_-$, and La_-. Each preserves

$$Q_{\mathrm{ch}} = N_s + N_+ - N_-. \qquad (20)$$

The tensor angular action is $2I_* Q_{\mathrm{ch}}$. The counter-rotating words change the unsigned excitation count by two while preserving this signed charge. No rotating-wave or photon-number truncation defines the receiving law. [1, §4]

6. The actual interacting clock

In a fixed lane define the bounded source amplitude and joint field form

$$Z = (D + iC)/d = \sigma_z + i\sigma_y,$$

$$z = \langle Z \rangle, \qquad w = \langle S(\phi_\times - i\phi_+) \rangle. \qquad (21)$$

The exact form equation and actual phase rate are

$$\dot{z} = -i\Delta z + 2gw,$$

$$\theta = -\arg z, \qquad \nu = \dot{\theta} = -\mathrm{Im}(\dot{z}/z). \qquad (22)$$

θ is the continuous lift with θ(0) = 0. A nonzero amplitude gives a continuously differentiable phase.

Split the frequencies at λ = 64Δ. The declared state is

$$\Psi_0 = \frac{\Phi_0 + \Phi_1}{\sqrt{2}} \otimes |0_s\rangle. \qquad (23)$$

Φ_0 and Φ_1 are the exact normalized interacting hard-field eigenbranches of signed charges zero and one. Their true gap is Ω > 0, and their relative phase gives $\langle \Phi_0 | Z | \Phi_1 \rangle = 2M$ with M > 0. This preparation includes the hard dressing and the full soft vacuum. [4, 5]

Use a comparison Hamiltonian with the same hard interaction, free soft field, even scalar, and exact scalar soft counterterm. In the same field realization its trajectory Ψ_{ref} has

$$h = Me^{-i\Omega\tau}, \qquad w_h = \frac{i(\Delta - \Omega)}{2g} h. \qquad (24)$$

Thus w_h is an actual reference expectation with zero mismatch. In particular ν(0) = Ω. The bare gap Δ, hard gap Ω, and actual rate ν retain distinct meanings.

More explicitly, $H_{ref} = \varepsilon e + H_h + H_{f,s} + c_s$, with the exact soft counterterm $c_s = 2g^2 \int_0^\lambda |u_R|^2/\omega \, d\omega$. Charge selection leaves the single transition in (24). Applying the hard-field clock equation to h then gives its stated w_h. Both propagators keep the same even scalar and exact counterterm, so the comparison retains their common phases.

6.1 A propagated centered bound

Set δz = z − h and δw = w − w_h. Subtracting the exact equations gives

$$\dot{z} + i\Omega z = -i(\Delta - \Omega)\delta z + 2g\delta w. \qquad (25)$$

The retained all-order soft estimate and energy bound are

$$|\delta z| \le E(\tau) = e^{4g^2 \nu_s^{\mathrm{up}} \tau^2} - 1,$$

$$\nu_s^{\mathrm{up}} = \frac{R^4 \lambda^6}{675\kappa\pi},$$

$$F_T = \left[\sqrt{c} + \sqrt{K_{\mathrm{prep}} + \varepsilon E(T)}\right]^2. \qquad (26)$$

K_{prep} is the retained upper bound on ⟨K⟩ in the exact state (23), including its full soft vacuum counterterm. The inherited all-time energy estimate gives $\langle K \rangle_\tau \le K_{prep} + \varepsilon E(\tau)$. Both actual and reference free energies are consequently at most F_T on [0,T]. For B = S(φ× − iφ+), the weighted insertion and full propagator satisfy

$$\|A^{-1/2} B A^{-1/2}\| \le \sqrt{2\mu},$$

$$\|A^{1/2} e^{-i\tau H} \eta\| \le 2\|A^{1/2}\eta\|. \qquad (27)$$

The second inequality uses the checked condition 2ε + 3c + 1/2 ≤ 2. Duhamel propagation of the soft interaction from the common initial vector then gives the achieved bound

$$\sup_{0 \le \tau \le T} |\delta w(\tau)| \le W_D(T),$$

$$W_D(T) = 8|g|T\sqrt{\mu \nu_s^{\mathrm{up}} (1 + F_T)(1 + F_T + \lambda)}. \qquad (28)$$

The comparison forcing creates a soft one-photon vector. Propagation is under the complete H and retains all photon sectors.

Proof of the propagated estimate. Put $H_{red} = \varepsilon S + K$ and $B_0 = H_{red} + (\varepsilon + c + 1/2)I$. The positive square form gives $H_f \le 2K + 2cI$ and $K \le 2H_f + 2cI$. The checked shift condition yields $A \le 2B_0 \le 4A$, so conservation of the B_0 form norm gives (27).

For $V_s = H - H_{ref}$, orthogonality of the two soft polarization insertions gives

$$\|A^{1/2} V_s \Psi_{ref}\|^2 \le 2g^2 \nu_s^{\mathrm{up}} (1 + F_T + \lambda).$$

Duhamel bounds the state difference in the energy form norm by 2T times this square root. The weighted B pairing contributes $2\sqrt{[2\mu(1 + F_T)]}$, yielding (28). [6]

6.2 Six uniform positive phase rates

For 0 < Ω ≤ Δ, suppose E(T) < M. Equations (22) and (25) imply

$$|\nu - \Omega| \le \frac{(\Delta - \Omega)E(T) + 2|g|W_D(T)}{M - E(T)}. \qquad (29)$$

The sufficient positivity condition simplifies, with the same true Ω throughout, to

$$2|g|W_D(T) + \Delta E(T) < \Omega M. \qquad (30)$$

Native outer evaluation satisfies (30) and the positive amplitude condition in every lane for

$$0 \le \tau \le T_0 = \frac{2\pi}{\Delta} = 573049\pi. \qquad (31)$$

Lane	Lower $10^6\nu$	Upper $10^6\nu$
01	3.490102929541	3.490102934505
02	3.490102893142	3.490102914462
03	3.490095588823	3.490098892473
12	3.490102927880	3.490102933590
13	3.490096147687	3.490099200196
23	3.490096722560	3.490099516734

Table 1. Uniform actual rates per normalized model time. Endpoints are rounded outward to twelve decimal places after multiplication by 10^6. Exact rational outer endpoints remain in the native certificate. [6]

All upper endpoints lie below Δ ≈ 3.49010294058623 × 10^{-6}. T_0 is one bare cycle; θ(T_0) is less than 2π. Every actual phase is strictly increasing on this interval and has an instantaneous inverse on its image. W_D upper bounds range from 1.229714 × 10^{-11} to 3.172702 × 10^{-10}.

For each certified lower rate ℓ > 0,

$$\theta(\tau_2) - \theta(\tau_1) \ge \ell(\tau_2 - \tau_1).$$

The inverse is continuously differentiable and Lipschitz with constant at most 1/ℓ. This is a uniform instantaneous statement, obtained by bounding the joint dynamics rather than differentiating a phase-error envelope.

6.3 Worked example: lane 01

The source-derived sign is $d_{01} < 0$ and hence $g_{01} < 0$. Representative values are

Quantity	Approximate value
d_{01}	−54.6776910663982
g_{01}	−9.54153851876510 × 10^{-5}
εe_{01}	0.917631085332839
c_{01}	2.78337517524759 × 10^{-8}

The complete source branch energies are approximately 0.917629340281369 and 0.917632830384310. Their midpoint is the retained even source energy, and their separation is Δ.

For the declared hard preparation, certified outer displays give

$$0.9999999987538952 < M < 0.9999999987925324,$$

$$3.4901029318881649 < 10^6\Omega < 3.4901029321578596. \qquad (32)$$

At τ = 0, z = M and the signed correlations are

$$\langle S\phi_\times \rangle = 0, \qquad \langle S\phi_+ \rangle = -\frac{(\Delta - \Omega)M}{2g} > 0. \qquad (33)$$

The plus correlation lies between 4.4166740 × 10^{-11} and 4.5580007 × 10^{-11}. It carries the initial clock shift from Δ to Ω. Replacing joint expectations by products of marginal means would change this equation.

The completed Duck campaign evaluated 300 native operations with 1,122 passing native checks. A separate checker passed 23,804 checks over all 300 new outputs and 134 matrix readouts. These are computational verification counts; the continuum argument is supplied analytically. The weighted-form, reference, and Duhamel construction is a Codex contribution developed within this collaboration, receiving the original source rules and retained hard preparation. [6]

7. The action in clock phase

Uniform positivity permits $d\tau = d\theta/\nu$ throughout (31). Every energy contribution receives the same actual inverse rate. For the canonical field form (14), $\dot{\phi} = \nu\partial_\theta\phi$, so the kinetic term also carries the velocity factor. The coefficient family in the lane action is

Contribution	Coefficient in θ
Field kinetic	ν/5
Field radial gradient	−1/(5ν)
Field angular term	−6/(5ν)
Even source energy	−εe/ν
Source exchange	−ε/ν
Boundary coupling	+g/ν
Both counterterms	−c/ν

Table 2. Coefficients inside the action divided by I_*. Field rows multiply their radial integrals; source and boundary rows multiply the corresponding joint expectations in (13). Field rows use (14); (11) already contains the full source-field Berry form. The full counterterm is $c = g^2/(2R)$.

The Berry contribution is the unchanged one-form

$$\frac{i}{2}(\langle\Psi|d\Psi\rangle - \langle d\Psi|\Psi\rangle). \qquad (34)$$

All entries depend on one underlying $\nu(\tau)$. Their interval images are dependent quantities, rather than independently selectable coefficients. The local inverse $1/\nu$ differs from the finite-window mean inverse $\tau/\theta(\tau)$. Likewise, a conserved energy's dynamical action $I_*E\tau$ differs from the full variational action (11), which retains its Berry term.

Equivalently, the complete joint-state action in this coordinate is

$$\frac{S}{I_*} = \int d\theta \left[\mathcal{B}_\theta - \frac{\langle H\rangle}{\nu}\right]. \qquad (35)$$

Here $\mathcal{B}_\theta d\theta$ is (34), and H is the full Hamiltonian (12). The even source energy and scalar lane counterterm remain in this expression even though they commute with Z and drop out of its clock equation. The same phase change therefore preserves the source, field, and interaction accounts together.

8. Chapter conclusion

The signed source construction, complete joint action, continuum form realization, and six uniform clock certificates close the Chapter I source-to-clock argument for the stated receiving law and preparation. **The test result suggests strong contact with the concept.** The centered target in the refinement note [2] is now supplied by the achieved bound (28) and its native evaluation. The historical derivations remain preserved in their original execution records.

The next source interface is the occupied-path relation between the original ledger $\hbar(d\gamma_W + 3d\chi)$ and the quantum generator $H_{pair}/2$. That work must carry the original common phase, the commuting even operator, and the preparation class. The six lane clocks established here provide the completed receiving calculation for that continuation. Radial geometry and entropy-area results belong to their later chapters. [3]

Source register

[1] Brady, with ChatGPT and Codex. *Original Write histories and a coupled source-field action*, v1.1, 15 October 2026. Owner-supplied manuscript.

[2] Brady, with ChatGPT and Codex. *A Reference-Centered Clock Positivity Condition*, 15 October 2026. Owner-supplied refinement note.

[3] *SAM Volume I Chapter Map and Research Plan*. Owner-supplied roadmap; its earlier manuscript references are superseded here by [1] and [6].

[4] *Original Write source coupling and six clocks*, H002628. Completed source campaign, 15 October 2026.

[5] *Resolved interacting clock*, retained source and hard-branch derivation, samland_resolved_clock1.

[6] *Chapter I lane01 and six uniform correlated clocks*, H002630, 15 October 2026; continuum, reference, and weighted-propagation derivations with native certificates. Exact source paths and hashes are retained with this draft.